Euler, the Basel sum, the Euler product, and coprime probability

Notes from Carlos' lessons

Source lessons: 06-07 (Euler: $\sum 1/n^2$), 06-12 (probability two random integers are coprime).

1 The Basel problem: $\displaystyle\sum_{n\geq 1}\frac{1}{n^2}=\frac{\pi^2}{6}$ (Euler, 1734)

Euler's idea (board). Let $f(x)=\dfrac{\sin x}{x}$ (with $f(0):=1$). Its roots are $\pm\pi,\pm 2\pi,\pm 3\pi,\dots$. Treat f as an "infinite polynomial" with those roots and constant term 1:

$$\frac{\sin x}{x}=\Big(1-\frac{x}{\pi}\Big)\Big(1+\frac{x}{\pi}\Big)\Big(1-\frac{x}{2\pi}\Big)\Big(1+\frac{x}{2\pi}\Big)\cdots=\prod_{n\geq 1}\Big(1-\frac{x^2}{n^2\pi^2}\Big).$$

Compare the coefficients of x^2.

- From the Taylor series $\sin x=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots$ (valid for all real x): $\frac{\sin x}{x}=1-\frac{x^2}{3!}+\frac{x^4}{5!}-\cdots$, so $[x^2]\frac{\sin x}{x}=-\frac{1}{6}$.
- From the product, the x^2 coefficient is $-\frac{1}{\pi^2}(1+\frac{1}{2^2}+\frac{1}{3^2}+\cdots)$, since only one factor can contribute its x^2 term.

Equating, $-\frac{1}{6}=-\frac{1}{\pi^2}\sum\frac{1}{n^2}$, so

$$\boxed{\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}}.$$

Caveat. This is Euler's original *heuristic*: a polynomial is determined by its roots and its value at 0, but an entire function need not be. It is made rigorous by the Weierstrass/Hadamard factorization, or by other proofs such as the Fourier-series one; the board treats it intuitively, as Euler did.

Zeta notation on the board: $\zeta(s)=\sum_{n\geq 1}n^{-s}$. Then $\zeta(1)=+\infty$ (harmonic series); $\zeta(r)<\infty$ for every $r>1$; $\zeta(\frac{1}{2})=\infty$; $\zeta(2)=\pi^2/6$; $\zeta(2k)$ is a rational multiple of π^{2k}; and $\zeta(3)=\sum n^{-3}$ is *not* of that form (no closed form in π).

2 The Euler product

Theorem 1. *For $s > 1$, $\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}$.*

Proof. For $|y| < 1$, $\sum_{k \geq 0} y^k = \frac{1}{1-y}$ (the geometric series; the board's "For $|x| < 1$, $\sum x^k = \frac{1}{1-x}$"). Multiply these over the primes $p = 2, 3, 5, \ldots$ with $y = p^{-s}$:

$$\Big(\sum_{k \geq 0} 2^{-ks}\Big)\Big(\sum_{k \geq 0} 3^{-ks}\Big)\Big(\sum_{k \geq 0} 5^{-ks}\Big) \cdots = \sum_{(e_2, e_3, e_5, \ldots)} \left(2^{e_2} 3^{e_3} 5^{e_5} \cdots\right)^{-s}.$$

By unique prime factorization every $n \geq 1$, $n = p_1^{a_1} \cdots p_\ell^{a_\ell}$, appears exactly once as $2^{e_2} 3^{e_3} \cdots$. The board expands the first two factors by hand: $(1 + \frac{1}{2^s} + \frac{1}{4^s} + \cdots)(1 + \frac{1}{3^s} + \frac{1}{9^s} + \cdots)$ contains $\frac{1}{n^s}$ for all n whose only prime factors are $2, 3$, and so on. So the product equals $\sum n^{-s}$ (for $s > 1$ everything converges absolutely, which justifies the rearrangement). □

At $s = 2$: $\zeta(2) = \Big(\frac{1}{1 - 2^{-2}}\Big)\Big(\frac{1}{1 - 3^{-2}}\Big)\Big(\frac{1}{1 - 5^{-2}}\Big) \cdots = \frac{\pi^2}{6}$ (the board's circled result), and $\frac{\pi^2}{6} \notin \mathbb{Q}$.

3 The probability that two random integers are coprime is $6/\pi^2$

The green note on the 06-07 board asks: take two integers a, b "at random" (the board admits that "random" is not yet defined). The probability that they are coprime is $\frac{6}{\pi^2} = \frac{1}{\zeta(2)}$.

3.1 Heuristic via the Euler product (06-12 board, right side)

1. For a random integer n, $P[p \mid n] = P[n \equiv 0 \pmod p] = \frac{1}{p}$.
2. a, b are coprime iff no prime divides both. $P[p \mid a \wedge p \mid b] = \frac{1}{p^2}$, so $P[p \nmid a \vee p \nmid b] = 1 - \frac{1}{p^2}$.
3. Treating different primes as independent,

$$P[\gcd(a, b) = 1] = \prod_p \Big(1 - \frac{1}{p^2}\Big) = \frac{1}{\prod_p (1 - p^{-2})^{-1}} = \frac{1}{\zeta(2)} = \frac{6}{\pi^2},$$

which is the board's boxed $\sum \frac{1}{n^2} = \frac{1}{(1 - \frac{1}{2^2})(1 - \frac{1}{3^2}) \cdots}$ (read through the reciprocal) combined with Section 1.

The independence step is informal: there is no uniform probability measure on $\mathbb{N}$, so the claim is rigorous only as a statement about limiting frequencies in $[1, n]^2$ (see the next subsection and the references).

3.2 Counting via Euler's totient (06-12 board, left side)

Let $P(n) = \dfrac{\big|\{(a, b) \in [n]^2 : \gcd(a, b) = 1\}\big|}{n^2}$. We want $\lim P(n)$.

Pair count. Write $\varphi(m) = |\{k \in [m] : \gcd(k,m) = 1\}| = \sum_{k=1}^{m} [\gcd(k,m) = 1]$. Pairs with $a \le b$ and gcd = 1: for each $b \ge 2$ there are $\varphi(b)$ choices of a (and $a = b$ is excluded since $\gcd(b, b) = b > 1$); for $b = 1$ there is the single pair $(1, 1)$. By the symmetry $a \leftrightarrow b$ the total is

$$1 + 2\sum_{b=2}^{n} \varphi(b) = 2\sum_{b=1}^{n} \varphi(b) - 1 \qquad (\text{since } \varphi(1) = 1).$$

The board shows exactly this $(\frac{1}{n^2}(1 + 2\sum_{b=2}^{n} \varphi(b)))$. Check for $n = 5$: $\varphi = 1, 1, 2, 2, 4$ has sum 10, giving 19, which matches a direct count of the coprime pairs in $[5]^2$.

Facts about φ (board).

- $\varphi(p^k) = p^k - p^{k-1} = p^{k-1}(p-1)$ (remove the multiples of p).
- $\varphi(ab) = \varphi(a)\varphi(b)$ if $\gcd(a, b) = 1$ (Chinese remainder theorem).
- Hence for $n = p_1^{a_1} \cdots p_\ell^{a_\ell}$: $\varphi(n) = n \prod_i (1 - \frac{1}{p_i})$.
- $\sum_{d|n} \varphi(d) = n$.

Proof of the last fact. Sort $\{1, \ldots, n\}$ by $g = \gcd(k, n)$. Those k with $\gcd(k, n) = n/d$ correspond to $k/(n/d) \in [d]$ coprime to d, so there are $\varphi(d)$ of them. Summing over $d \mid n$ counts all n numbers. □

The limit (my completion; the board stops after setting up $S(n) = \sum_{k \le n} \varphi(k)$). Möbius inversion of $\sum_{d|n} \varphi(d) = n$ gives $\varphi(k) = \sum_{d|k} \mu(d)\, \frac{k}{d}$, so

$$S(n) = \sum_{k \le n} \varphi(k) = \sum_{d \le n} \mu(d) \sum_{m \le n/d} m = \sum_{d \le n} \mu(d)\, \frac{\lfloor n/d \rfloor (\lfloor n/d \rfloor + 1)}{2} = \frac{n^2}{2} \sum_{d \ge 1} \frac{\mu(d)}{d^2} + O(n \log n),$$

and $\sum_d \mu(d) d^{-2} = \prod_p (1 - p^{-2}) = \frac{1}{\zeta(2)}$ (the Euler product again). So $S(n) \sim \frac{3}{\pi^2} n^2$ and $P(n) = \frac{2S(n)-1}{n^2} \to \frac{6}{\pi^2}$. Numerically, $S(2000)/2000^2 = 0.3041$ against $3/\pi^2 = 0.3040$.

Asides on these boards

- $1 + 2 + 3 + 4 + \cdots = -\frac{1}{12}$ appears (flagged) on the 06-12 board, with $\zeta(k) = S(\infty, -k) = \lim_{n \to \infty} S(n, -k)$ and $S(k, n) = \sum_{i \le n} i^k$: the regularized value $\zeta(-1) = -\frac{1}{12}$, not a convergent sum.
- The 06-12 board also lists derivative rules for formal series: $(A + B)' = A' + B'$, $(AB)' = A'B + AB'$, $(x_0, x_1, x_2, \ldots)' = (x_1, 2x_2, 3x_3, \ldots)$; and $\sum nx^n = \frac{x}{(1-x)^2}$, i.e. $(0, 1, 2, 3, \ldots)$; and the recurrence $a_{n+1} = 2a_n + n$, which is solved in the topic *Ordinary generating functions*.

Sources

- The probability that two random integers are coprime — Bureaux & Enriquez and On the probability that two random integers are coprime — Lei & Kadane: the precise meaning of "random integer" (no countably additive uniform measure on $\mathbb{N}$).

- Probability of Two Integers Being Coprime (cut-the-knot); Cushing your luck (The Aperiodical): $1/\zeta(2)$ and the Basel value.
- The totient/Möbius completion above is standard (e.g. Hardy & Wright, Thm. 331, as cited by Lei–Kadane); the numeric checks were run locally.